Ambery (Yu-Jung) Liang

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Research Interests

Safety and reliability of LLM agents, especially multi-agent systems in adversarial, medical, and long-horizon settings. I study the mechanisms that decide whether agents stay reliable.

Education

University of California San Diego, MS in Electrical & Computer Engineering — Sep. 2025 – Jun. 2027
Specialization: Machine Learning and Data Science

National Chung Hsing University, in Management of Information Systems — Sep. 2021 – Jun. 2025
Ranked 1st in class, Fall 2024

Publications

[1] **Yu-Jung Liang***, Yu-Shu Chen*. To Diffuse or Not to Diffuse: Matching Graph Retrieval to Memory Structure in LLM Agents. Lifelong Agent Workshop @ COLM 2026 (accepted, poster). Full version under review at ICLR 2027.

[2] **Yu-Jung Liang**. Teammate Awareness Makes a Coordination Signal Visible: A First Step Toward Detecting Collusion in Multi-Agent LLM Debate. WiML @ NeurIPS 2026 (accepted).

[3] **Yu-Jung Liang***, Yu-Shu Chen*. TagGraph: Tag-Augmented Graphs for Graph Retrieval of Agent Persistent Histories. NAACL 2027.**Under review**

[4] **Yu-Jung Liang**. Talking More Hurts: Insider Collusion Attack-Defense in Medical Multi-Agent Question-Answering Systems. Target: ICML 2027.**In preparation**

*Equal contribution.

Research Experience

Insider Agents in Multi-Agent Debate: Measurement, Mechanism, Defense — July 2026 – present
In collaboration with Prof. Carlee Joe-Wong (CMU ECE) · Papers [2], [4]

- Designed a black-box insider attack insider the multi-agent debate system in which two scripted agents argue for the same wrong option, to measure how far coordinated peers can pull honest agents without access to any model's weights. Across three published debate systems, two medical QA sets (MedMCQA, MedQA), and three open backbones (Qwen2.5-Coder-7B and 14B, Llama-3.1-8B), it moved 37.7% (MedLA), 41.3% (MDAgents), and 50.0% (ReConcile) of honest Coder-7B agents onto the target on MedMCQA. On MedQA, final answers moved for 48% (MDAgents) and 32% (MedLA); Llama-3.1-8B on ReConcile was hit as hard to 53%.
- Designed the full defense as a safety layer that acts only at the turn an agent is being pulled, so the rest of the debate runs untouched. Each honest turn has three steps: the agent answers as usual; if it changed its answer and a linear probe on its layer-24 activations flags the change as harmful, it answers again with the harmful direction subtracted from those activations; only the second answer reaches the other agents. The probe, its threshold, and the direction are fit separately for each system.
- Placed the defense during the debate, aiming for a dynamic model change: adjusting each honest agent's internal computation turn by turn at inference time, rather than retraining it or filtering votes afterward. It prevented up to 35.4% of honest agents' moves onto the insiders' target (23.2% on MDAgents, 34.5% on MedLA, 35.4% on ReConcile).
- Probed the residual stream at the answer token to locate where a harmful switch becomes readable. We discovered at layer 24, a linear probe separated harmful from helpful switches across insider targets (AUROC:0.76).
- Gated a re-ask mechanism on the probe. Re-ask the same question to agent so that only harmful switches get corrected and the helpful ones that make debate useful are kept. When the probe flags a switch, the agent answers again with the harmful direction subtracted from its activations. This cut honest agents moving to the target by 13.0% (MedLA), 9.6% (MDAgents), and 17.7% (ReConcile) percentage points, and raised ReConcile's accuracy from 16% to 29%.

Position Bias in Medical Multi-Agent LLM Systems — May 2026 – Sep. 2026

- Designed a black-box attack that reorders answer options without changing their content and ran it on four medical multi-agent systems (self-designed 5-agent majority vote, MDAgents basic and intermediate version, MedAgents) on one backbone, to test whether deliberation removes a single model's position bias. Found that MDAgents' deliberation was nearly twice as vulnerable to prompt injection as majority voting (45% vs. 22% attack success), with no improvement in clean accuracy.
- Designed a counter-permutation defense that re-asks under k random option orders and votes on option content, because it needs no retraining and mixing over orders cancels any fixed reordering. At k=3 it raised the majority vote's attacked accuracy up to 6%, but on MDAgents intermediate k=7 trailed k=3 and 5 (35% vs. 45%, n=20) because larger k widens content-vote ties, so k has to be set per system.

Graph-Structured Memory and Retrieval for LLM Agents — Apr. 2026 – Oct. 2026

Advisor: Prof. Pengtao Xie · Papers [1], [3]

- Developed a structured long-term memory extraction pipeline based on 5W (Who, What, When, Where, Why), with entity canonicalization and graph-based storage. Integrated the system into OpenClaw as a graph-memory plugin using TypeScript and Python NetworkX, achieving 79.7 Recall@5 on LongMemEval with a simple dense retrieval baseline.
- Built a controlled retrieval evaluation framework across LongMemEval, ATANT, LoCoMo, and MemBench, isolating retrieval performance by holding extracted memories fixed across methods. Dense retrieval significantly outperformed the original tag-graph retriever (Recall@5: 82.5 vs. 41.8; paired bootstrap, ($p \leq 0.001$)), identifying retrieval algorithms as a key performance bottleneck.
- Improved graph-based lexical retrieval through vocabulary normalization, stopword filtering, and TF-IDF weighting, increasing Recall@1 from 12.1 to 19.6 (+62.6%) and Recall@5 from 41.8 to 53.7. Further evaluated raw-conversation BM25 as a no-extraction baseline, revealing that structured memory extraction provides benchmark-dependent benefits and that missing information at ingestion remains a major source of retrieval failure.
- Investigated graph diffusion under varying memory-store sizes by introducing up to 36 distractor sessions into LongMemEval, observing up to +0.58 MRR improvement. Redesigned TagGraph with a shared hierarchical 5W+Extra ontology and a hybrid retrieval-and-ranking architecture, achieving 42.7%, 57.1%, and 65.7% end-to-end QA accuracy at 2K, 4K, and 8K context tokens, respectively.

AI Engineer, Far Eastern Memorial Hospital, AI Research Team (Remote, Taiwan) Jun. 2025 – May 2026

Advisor: Dr. Fang-Ming Hung · Multi-agent clinical decision support for cardiology

- Developed a multi-agent clinical decision-support system for cardiology using LangGraph, Qwen (DashScope), and Neo4j, implementing three LLM-powered workflows for ICD-10 diagnosis inference, treatment recommendation, and medication recommendation from free-text admission notes, with physician-in-the-loop review.
- Designed and implemented a patient-centered temporal knowledge graph in Neo4j, modeling longitudinal patient records through Patient–Visit relationships and sequential temporal edges (NEXT), with visit-level diagnosis, medication, and treatment event nodes to represent evolving clinical histories.
- Designed a human-in-the-loop clinical workflow enabling physicians to review, reorder, and revise model-generated recommendations. Explored a dual-source context-fusion approach combining patient-specific historical records with analogous clinical cases to ground LLM-generated recommendations, and designed a physician-feedback mechanism for incorporating clinical corrections as reusable knowledge-graph guidelines. Proposed a pre/post evaluation framework using CBI, PSS-10, DFS, and NASA-TLX to assess potential impacts on clinician burnout, perceived stress, decision fatigue, and cognitive workload.

Undergraduate Researcher, National Chung Hsing University (Taiwan) Sep. 2023 – Mar. 2025

Advisor: Prof. Meng-Hsiun Tsai · Funded by the National Science Council undergraduate research program

- Developed a hybrid breast-ultrasound lesion analysis pipeline integrating U-Net segmentation for lesion boundary delineation and YOLOv8 for lesion localization, achieving 95% precision and 92% recall across 3,000+ labeled scans. Conducted prediction-map-based error analysis to identify low-contrast tumors as a major source of model failures.

Selected Projects

Do LLM Embeddings Outperform Traditional Models in Diabetic Comorbidity Prediction? Jan. 2026 – Mar. 2026

- Formulated a longitudinal next-comorbidity prediction task using MIMIC-IV data from 27,621 patients with type 2 diabetes, predicting the next previously unseen comorbidity among 26 clinically robust classes. Systematically compared frozen Qwen3-1.7B embeddings, conventional machine learning models, and BEHRT, finding that general-purpose LLM embeddings (MRR: 0.38) underperformed XGBoost (MRR: 0.42).
- Expanded BEHRT's diagnosis vocabulary from 26 target comorbidity tokens to approximately 2,400 frequent ICD-9/10 codes derived from 1.26 million diagnosis records, preserving the temporal order of patients' longitudinal diagnosis histories. This richer clinical representation increased MRR from 0.43 to 0.62.
- Developed a hybrid Transformer–XGBoost framework by extracting [CLS] representations from pretrained and task-fine-tuned EHR Transformers, concatenating them with structured clinical features, and training XGBoost as a downstream predictor. Evaluated 21 model configurations and achieved 40–50% relative MRR improvements over corresponding MLP-head variants, with pretrained BEHRT reaching 0.66 MRR and 85.9% Top-5 accuracy.

Prompt Injection Vulnerabilities in Medical Multi-Agent Debate Systems Oct. 2026 – present

- Designed an adversarial evaluation framework for open-form medical multi-agent systems using the Medical Prompt Injection Benchmark (MPIB), examining how malicious instructions influence collective clinical reasoning and final medical decisions across three debate architectures: Multi-Agent Debate, MAD, and ReConcile.
- Investigated system-level robustness under varying numbers of compromised agents, evaluating how attack propagation, injection timing, and debate mechanisms affect the likelihood of incorrect or attacker-targeted medical decisions.

Evaluating the Evaluators: Are Memory Benchmarks Rewarding Generality? Oct. 2026 – present

- Designed item-level tests of whether each LoCoMo and LongMemEval question needs memory at all (question-only, oracle, and full-pipeline triage; shuffled-history ablations), because LongMemEval's "no shortcut" claim rests on co-occurrence with correct retrieval, not a closed-book baseline. IRT item analysis then finds which items decide method rankings and whether

the rankings survive reweighting.

Professional Experience

Data Analytics Intern, ASML (Taiwan) Jul. 2024 – Jun. 2025

- Automated appointment scheduling over 500+ weekly records (Python, SQL, Power BI), replacing about 100 hours of manual work per week; rule-based no-show detection let operations leads reallocate weekly capacity.

Honors and Awards

Academic Excellence Award, NCHU (ranked 1st, Fall 2024) · **3rd Place**, IM Capstone Presentation, NCHU (2024)

Technical Skills

ML & LLMs: PyTorch, Transformers, multi-agent LLMs, RAG, LangGraph, residual probes, activation steering, BEHRT
Methods: controlled ablations, paired bootstrap, difference-in-differences, pre-registered kill criteria, adaptive attacks
Systems: Python, TypeScript, SQL, Neo4j, NetworkX, Docker, Git, Linux, Kubernetes, SSH